Holistic Reasoning with Long-Context LMs:

A Benchmark for Database Operations on Massive Textual Data

(*ICLR 2025*)

Seiji Maekawa*, Hayate Iso* and Nikita Bhutani
Megagon Labs

Megagon Labs

Seiji MAEKAWA（前川政司）

2017 2019 2020 2023

Full-time

柔道

阪大修士

法人営業

NTT docomo

阪大博士

Research Scientist

Megagon Labs

USA

Part-time

京大学士

KYOTO UNIVERSITY FOUNDED 1897

1M

Hong Kong

2M

TU/e EINDHOVEN UNIVERSITY OF TECHNOLOGY

Netherlands

2M

hottolink

Remote

3M

Megagon Labs

USA

柔道

Research Topic

トピックモデル
With 単語ベクトル

人工グラフ生成＆クラスタリング

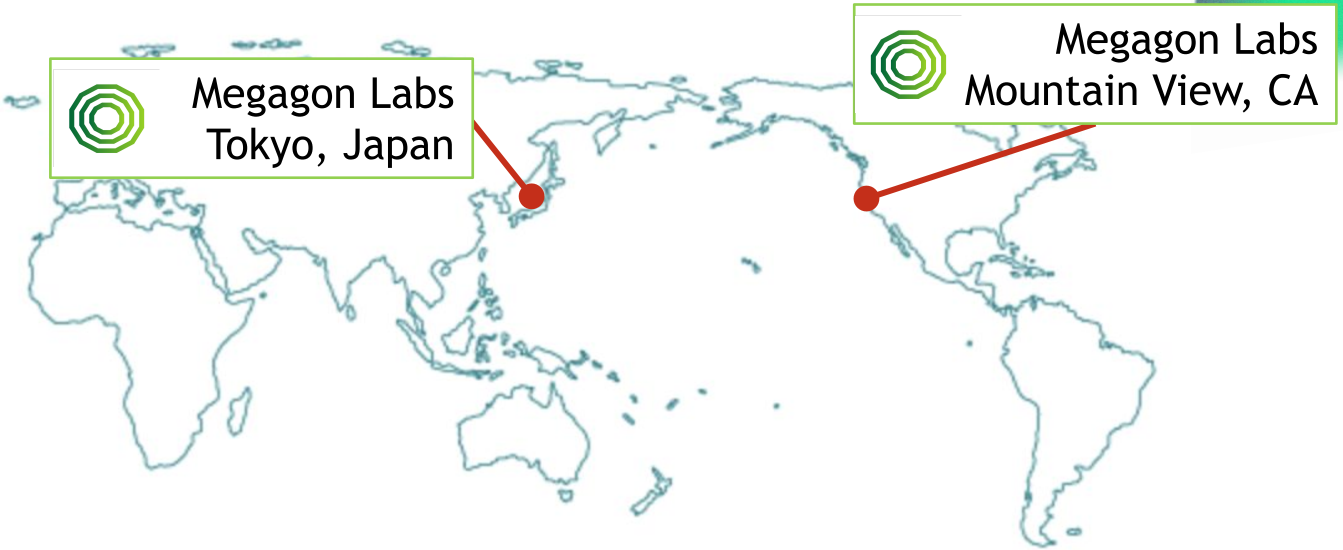
営業

GNNsのベンチマーク＆高速化

LLMの検索拡張 (RAG) ＆
長文脈能力の調査、改善

Megagon Labs

Where We Are



RECRUIT

Megagon Labs

indeed

glassdoor

CSI Companies

CHANDLER MACLEOD GROUP

unique

Staffmark Group

Quandoo

Relevant Recent Publications

NLP, LLM, & Machine Learning

- **Holistic Reasoning with Long-Context LMs: A Benchmark for Database Operations on Massive Textual Data - ** ICLR 2025
- **Multi-Conditional Ranking with Large Language Models - ** NAACL 2025
- **LLMs Are Not Intelligent Thinkers: Introducing Mathematical Topic Tree Benchmark for Comprehensive Evaluation of LLMs - ** NAACL 2025
- **From Single to Multi: How LLMs Hallucinate in Multi-Document Summarization - ** NAACL 2025 (Findings)
- **Natural Language Processing for Human Resources: A Survey - ** NAACL 2025 (Industry)
- **Evaluating Bias in LLMs for Job-Resume Matching: Gender, Race, and Education - ** NAACL 2025 (Industry)
- **AmbigNLG: Addressing Task Ambiguity in Instruction for NLG - ** EMNLP 2024
- **Characterizing Large Language Models as Rationalizers of Knowledge-intensive Tasks** - ACL 2024 (Findings)
- **Retrieval Helps or Hurts? A Deeper Dive into the Efficacy of Retrieval Augmentation to Language Models - ** NAACL 2024
- **Large language models sensitivity to the order of options in multiple-choice questions - ** NAACL 2024 (Findings)
- **Less is more for long document summary evaluation by LLMs** - EACL2024
- **XATU: A Fine-grained Instruction-based Benchmark for Explainable Text Updates** - COLING 2024

Relevant Recent Publications

Data Management & Integration

- **Orchestrating Agents and Data for Enterprise: A Blueprint Architecture for Compound AI -** DAIS @ ICDE 2025
- **Retrieval Augmented Generation in the Wild: A System 2 Perspective -** IEEE ICDE 2025
- **Watchog: A Light-weight Contrastive Learning based Framework for Table Understanding -** SIGMOD 2024
- **Demonstration of a Multi-agent Framework for Text to SQL Applications with Large Language Models -** CIKM 2024 (demo)
- **CMDBench: A Benchmark for Coarse-to-fine Multimodal Data Discovery in Compound AI Systems -** GUIDE-AI @ SIGMOD 2024
- **Fairness-aware Data Preparation for Entity Matching -** ICDE 2024
- **Investigation of Simple-but-Effective Architecture for Long-form Text Matching with Transformers –** DASFAA 2024

Relevant Recent Publications

Human-AI

- **VeriLA: A Human-Centered Evaluation Framework for Interpretable Verification of LLM Agent Failures -** HEAL @ CHI2025
- **Human-LLM Collaborative Annotation Through Effective Verification of LLM Labels -** CHI 2024
- **Human-LLM collaborative annotation framework -** EACL2024 Demo
    - MEGAnno+ Open Source Tool
- **Magneton - Towards Integrated, Extensible, and Reproducible Knowledge Graph Exploration** - CHI 2023
    - Open Source Tool
- **Weedle: Composable Dashboard for Data-centric NLP in Computational Notebooks -** Demo at TheWebConf 2023

Holistic Reasoning with Long-Context LMs:

A Benchmark for Database Operations on Massive Textual Data

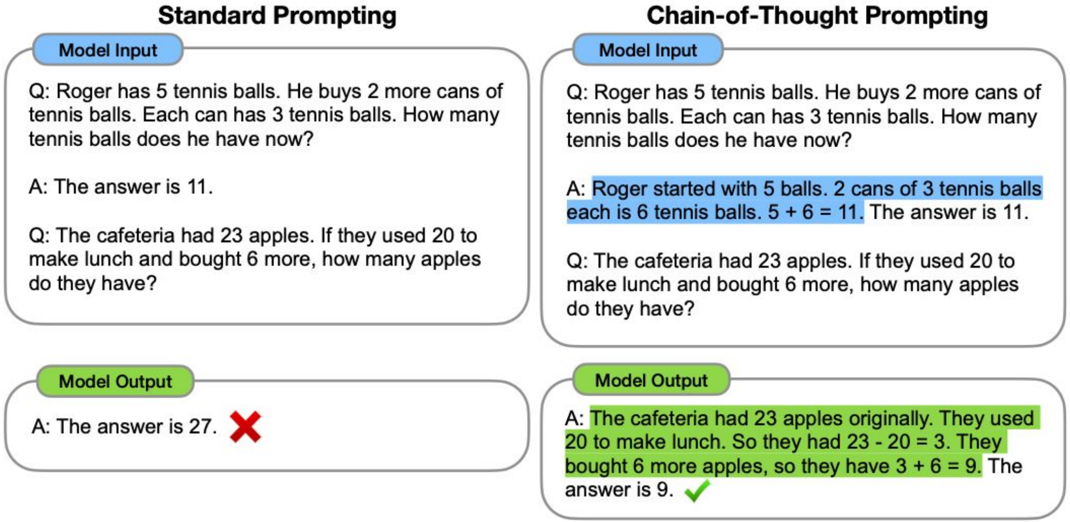
(*ICLR 2025*)

Seiji Maekawa*, Hayate Iso* and Nikita Bhutani
Megagon Labs

Megagon Labs

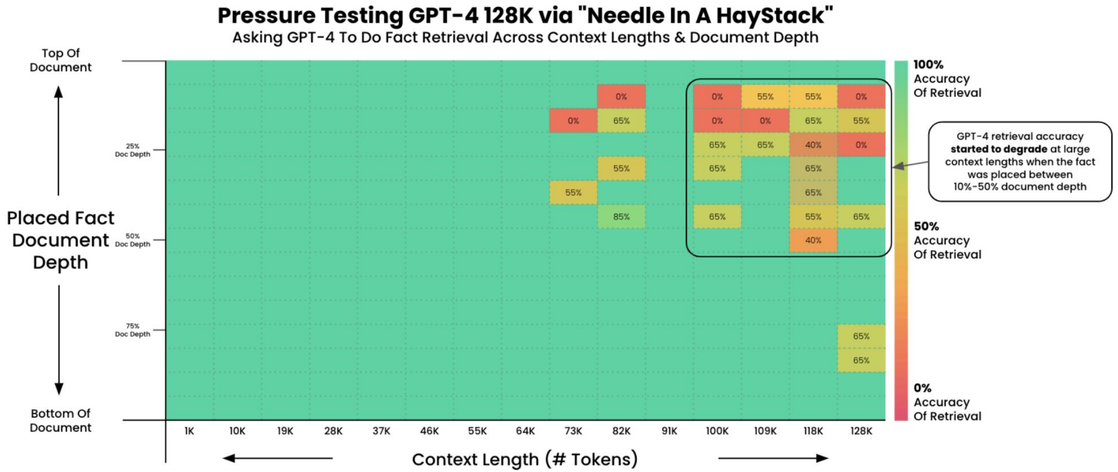
LLMsの推論能力

- 回答前に推論過程（理由づけ）を明示させる
    - いわゆる Chain-of-Thought（思考の連鎖）プロンプト
- 適応的な計算ステップを許容する
    - 難しい質問ほど LLM に「考える時間」を与える発想



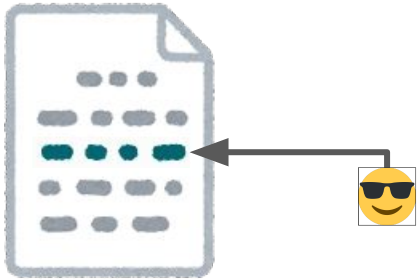
「長い文書」に対する推論は？

- コンテキスト長は急速に拡大中
  - 例: Llama-1 (2k) → Llama-2 (4k) → Llama-3 (8k) → Llama-3.1 (128k) → Llama-4 (10M)
- 既存評価は retrieval 中心
- 代表例「Needle-in-a-Haystack」テスト：
  - 指定フレーズを抜き出せるか を測るだけで「推論」は要らない

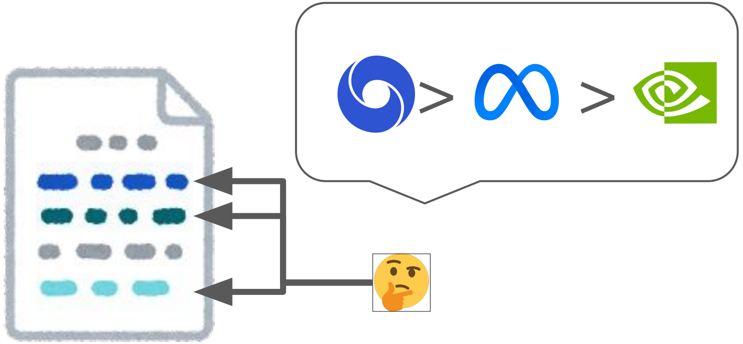


単純検索はできるが “全体を見渡す推論” は苦手

Simple Lookups
Q: What was NVIDIA’s revenue?



Holistic Reasoning
Q: Which company had the highest revenue?

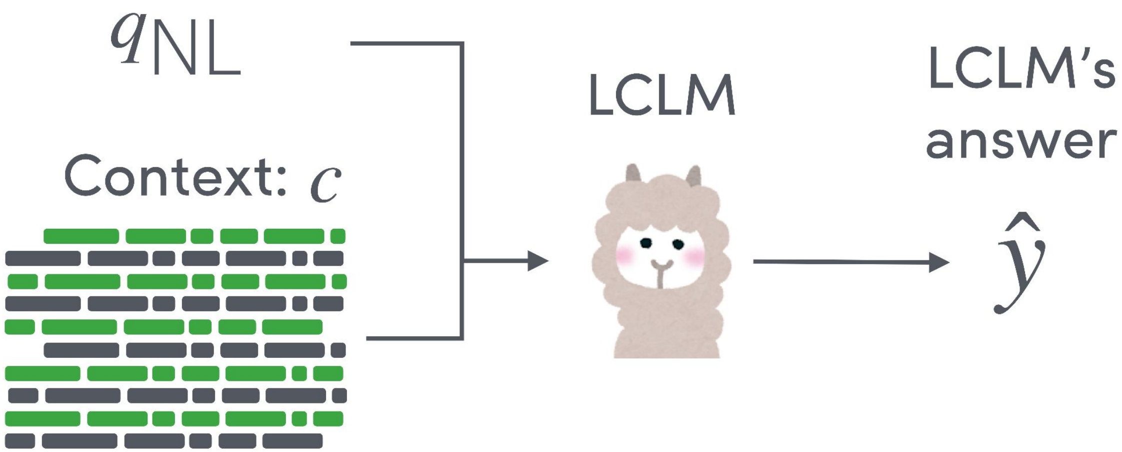


膨大なテキストを跨いで情報を集計・比較する能力が不足

今日お話しすること

研究概要: Holistic Reasoning with Long-context LMs (ICLR2025)

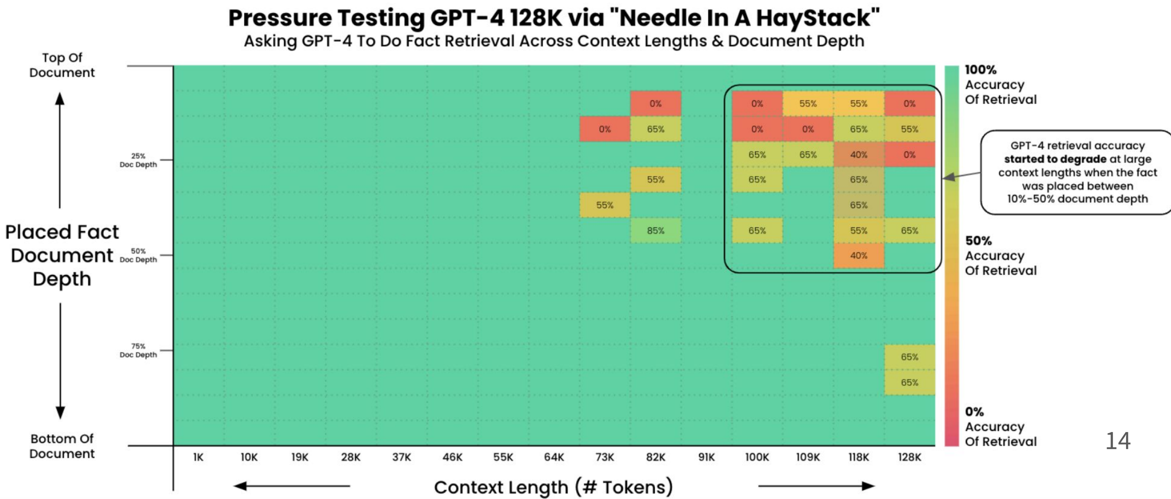
- 長大コンテキスト LLM が データベース的推論 をどこまで行えるか検証
- 入力も推論経路も “長い” 状態で評価



Needle-in-haystack test (Kamradt, 2023)

- ターゲット:
  - Q: サンフランシスコで最高の過ごし方は？
  - A: 晴れた日にドルレス公園でサンドイッチを食べること。
- 文脈:
  - ポール・グラハムのエッセイ

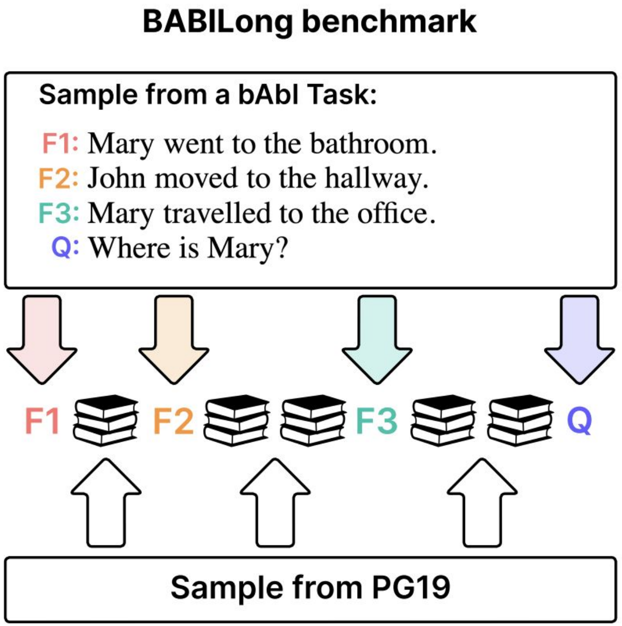
- 課題:
  - 検索のみ。推論は不要。



BABILong: Testing the Limits of LLMs with Long Context Reasoning-in-a-Haystack (Kuratov+, NeurIPS 2024)

- ターゲット:
  - bAbI task; 単純な推論タスク
- 文脈:
  - PG19 (book corpus)
- 課題 :
  - 論理難易度は低い
  - 文脈が完全に無関係な場合すらある

a)



既存の長文脈ベンチマークとの比較

Benchmark	Variable context len.	Variable info. amount	Variable info. position	Reasoning over 100+ info. pieces	Verifiable answers
Needle-in-a-haystack (Kamradt, 2023)	✓	✗	✓	✗	✓
Multi Needles-in-a-haystack (Kamradt, 2023)	✓	✓	✓	✗	✓
En.QA/En.MC (Zhang et al., 2024)	✗	✗	✗	✗	✗ human-anno.
BABILONG (Kuratov et al., 2024)	✓	✗	✓	✗	✓
NOCHA (Karpinska et al., 2024)	✗	✗	✗	✗	✗ human-anno.
FLENQA (Levy et al., 2024)	✓	✗	✓	✗	✓
HOLOBENCH (Ours)	✓	✓	✓	✓	✓

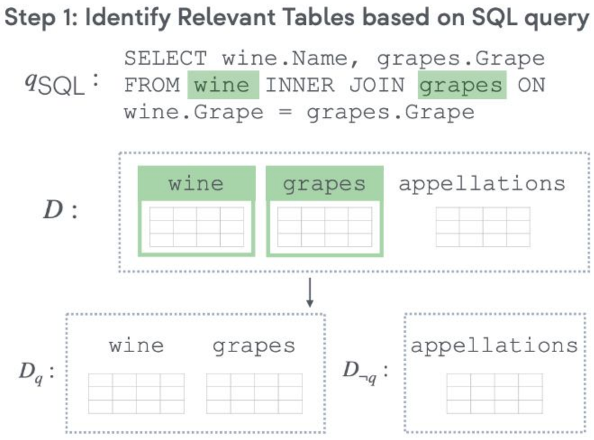
HoloBench

- 長文上でデータベース操作
  - 例 : MAX, COUNT DISTINCT, GROUP BY 相当を自然言語で質問
  - 非構造テキスト ↔️ SQL を自動で対応付け
  - 生成回答を実際の SQL 実行結果と比較することで自動採点

https://github.com/megagonlabs/holobench

HoloBench

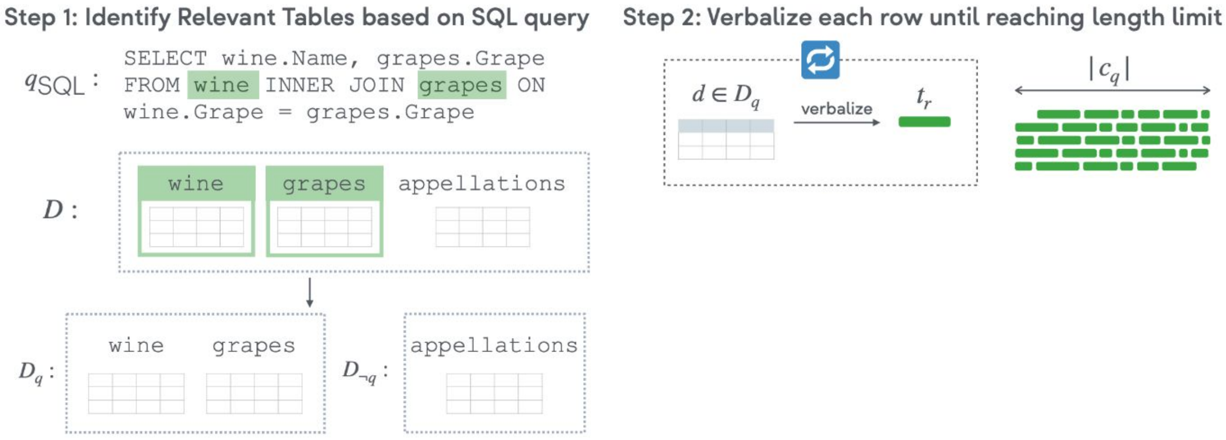
- 長文上でデータベース操作
  - 例 : MAX, COUNT DISTINCT, GROUP BY 相当を自然言語で質問
  - 非構造テキスト ↔️ SQL を自動で対応付け
  - 生成回答を実際の SQL 実行結果と比較することで自動採点



https://github.com/megagonlabs/holobench

HoloBench

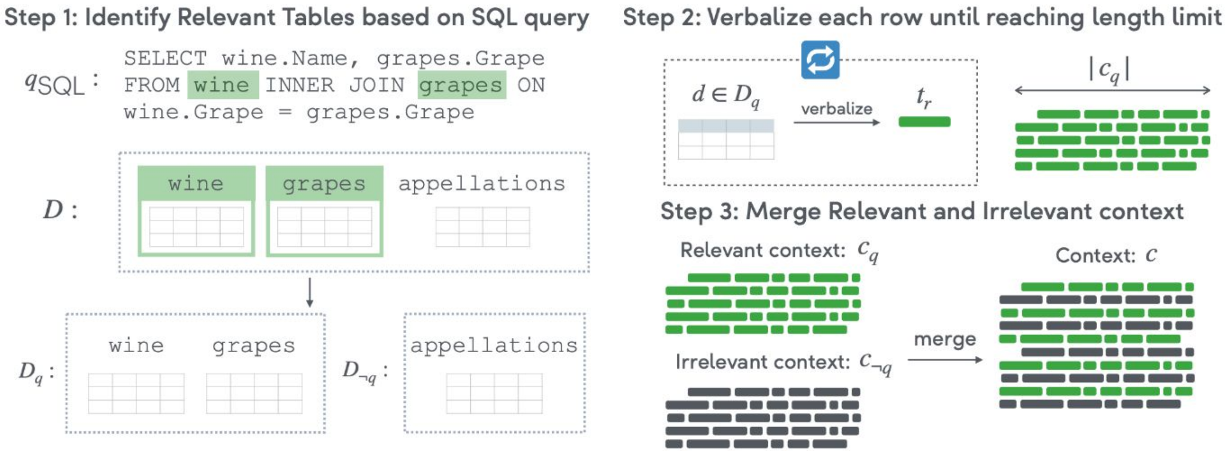
- 長文上でデータベース操作
  - 例 : MAX, COUNT DISTINCT, GROUP BY 相当を自然言語で質問
  - 非構造テキスト ↔️ SQL を自動で対応付け
  - 生成回答を実際の SQL 実行結果と比較することで自動採点



https://github.com/megagonlabs/holobench

HoloBench

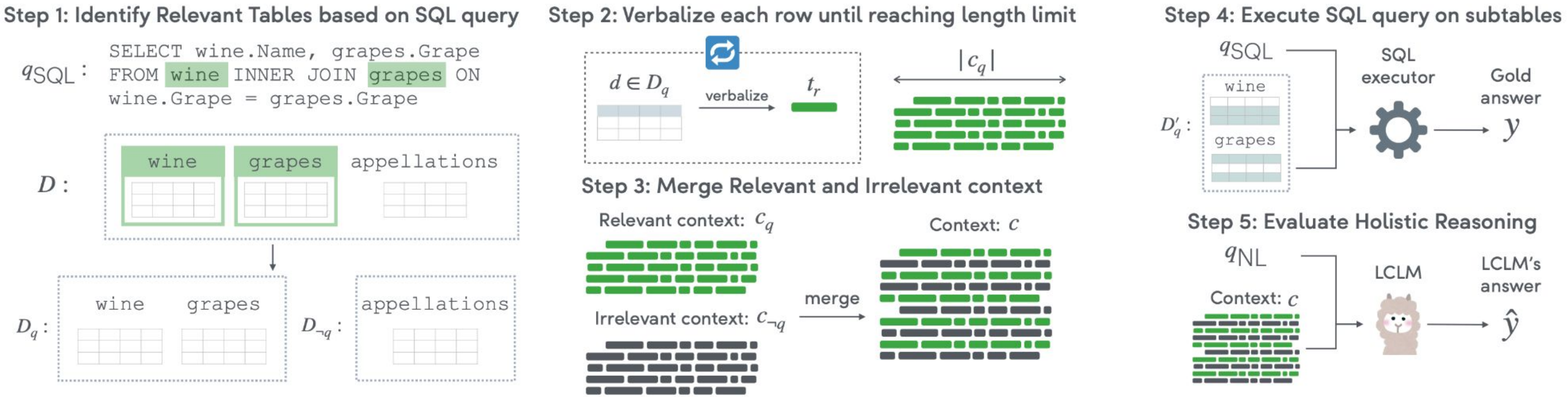
- 長文上でデータベース操作
  - 例 : MAX, COUNT DISTINCT, GROUP BY 相当を自然言語で質問
  - 非構造テキスト ↔️ SQL を自動で対応付け
  - 生成回答を実際の SQL 実行結果と比較することで自動採点



https://github.com/megagonlabs/holobench

HoloBench

- 長文上でデータベース操作
  - 例 : MAX, COUNT DISTINCT, GROUP BY 相当を自然言語で質問
  - 非構造テキスト ↔️ SQL を自動で対応付け
  - 生成回答を実際の SQL 実行結果と比較することで自動採点



https://github.com/megagonlabs/holobench

変換例（自然言語 ↔️ SQL）

```sql
-- SQL Query
SELECT destination, COUNT(DISTINCT source)
FROM flights
GROUP BY destination
ORDER BY COUNT(DISTINCT source) DESC
LIMIT 1
```

Natural Language Query: "Which airport receives flights from the most distinct source airports? Include the number of source airports."
（「最も多くの出発空港から便を受け入れている空港と、その空港数は？」）

Context: "Flight 101 departs from JFK and arrives at LAX..." [... many flight descriptions ...]
（「フライト 101 は JFK 発 LAX 着…」（数千行続く））

Details in the paper :)

実験設定

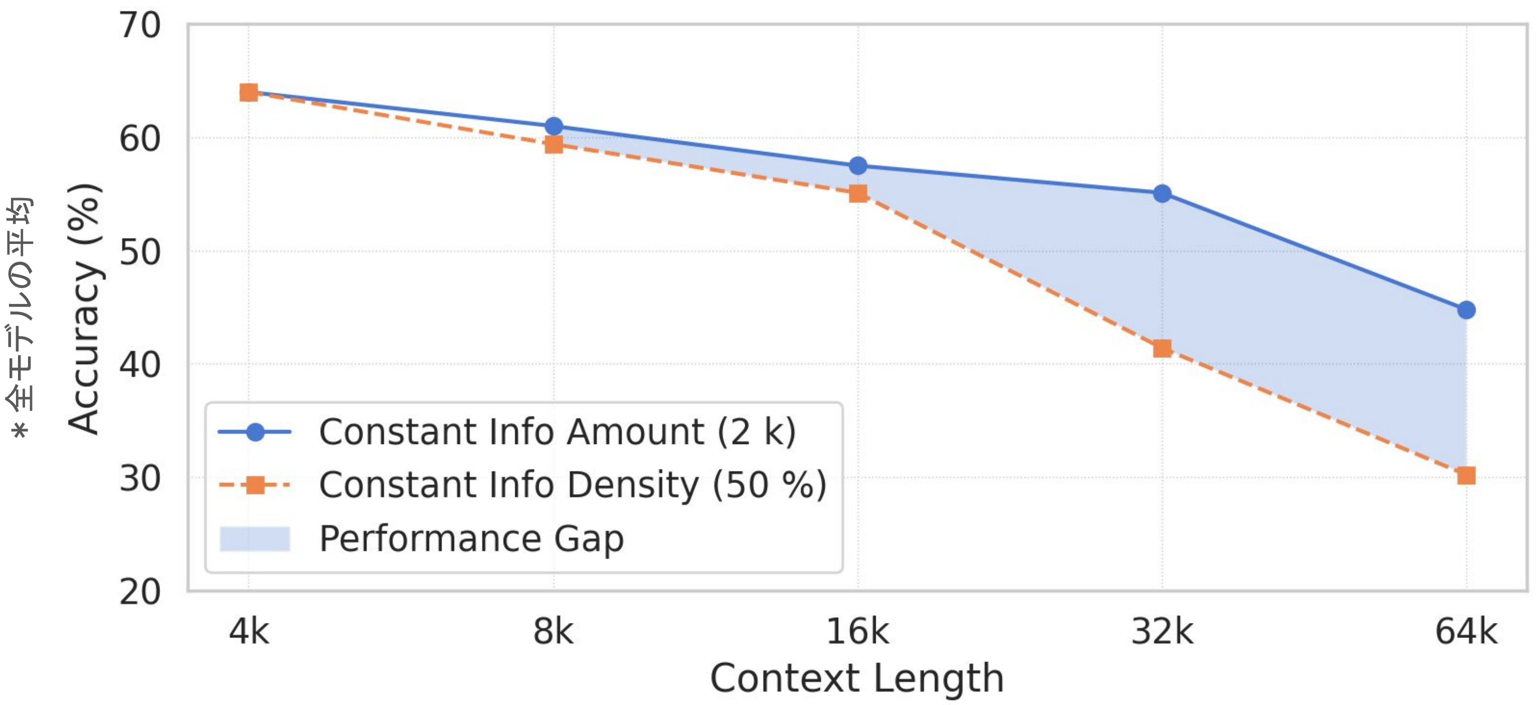
モデル

- Llama 3.1 (8B/70B/405B)
- GPT-4o/GPT-4o-mini, o1-mini
- Claude 3.5 Sonnet
- Gemini 1.5 Flash/Pro

変数

- 文脈長: 4k-64k (tokens)
- 情報密度: Fixed info (2k tokens), 50% information density
- 情報ポジション: Uniform, Beginning, Middle, End, Bimodal

情報量の増加 はコンテキスト長よりモデル性能に影響大



詳細な性能分析

LMs	Constant Information Amount (2k)								Constant Information Density (50%)							
	4k	8k	16k	32k	64k	Avg.	wAvg. (inc)	wAvg. (dec)	4k	8k	16k	32k	64k	Avg.	wAvg. (inc)	wAvg. (dec)
Llama-3.1-405b	64.0	61.0	57.5	55.1	44.8	56.5	53.5	59.4	64.0	59.4	55.1	41.4	30.2	50.0	44.3	55.7
GPT-4o	62.5	52.2	53.6	59.4	52.0	55.9	55.0	56.8	62.5	57.5	45.3	43.3	37.5	49.2	45.0	53.5
o1-mini	63.5	55.8	48.8	47.6	40.3	51.2	47.5	54.8	63.5	51.2	45.8	34.4	29.2	44.8	39.1	50.5
Claude-3.5-Sonnet	49.3	52.2	46.4	42.3	41.9	46.4	44.8	48.1	49.3	50.7	47.2	42.1	34.7	44.8	42.3	47.3
Llama-3.1-70b	60.5	51.2	41.1	42.2	33.6	45.7	41.5	49.9	60.5	53.7	45.9	39.0	28.6	45.5	40.3	50.8
GPT-4o-mini	51.5	50.1	43.1	43.2	38.7	45.3	43.2	47.5	51.5	45.6	37.8	35.0	26.3	39.2	35.2	43.3
Gemini-1.5 Pro	45.6	43.4	32.8	39.2	42.2	40.6	39.9	41.4	45.6	35.1	31.9	30.0	31.5	34.8	32.6	37.0
Gemini-1.5 Flash	36.8	32.0	34.3	27.2	26.8	31.4	29.8	33.1	36.8	30.7	25.8	27.7	20.5	28.3	25.9	30.7
Llama-3.1-8b	39.1	27.4	28.7	9.8	10.0	23.0	18.0	28.1	39.1	34.7	21.0	6.6	1.1	20.5	13.5	27.4

詳細な性能分析

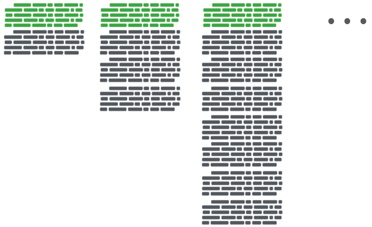
Llama-3.1-405b が“全体を見渡す推論”で良いパフォーマンス

LMs	Constant Information Amount (2k)								Constant Information Density (50%)							
	4k	8k	16k	32k	64k	Avg.	wAvg. (inc)	wAvg. (dec)	4k	8k	16k	32k	64k	Avg.	wAvg. (inc)	wAvg. (dec)
Llama-3.1-405b	64.0	61.0	57.5	55.1	44.8	56.5	53.5	59.4	64.0	59.4	55.1	41.4	30.2	50.0	44.3	55.7
GPT-4o	62.5	52.2	53.6	59.4	52.0	55.9	55.0	56.8	62.5	57.5	45.3	43.3	37.5	49.2	45.0	53.5
o1-mini	63.5	55.8	48.8	47.6	40.3	51.2	47.5	54.8	63.5	51.2	45.8	34.4	29.2	44.8	39.1	50.5
Claude-3.5-Sonnet	49.3	52.2	46.4	42.3	41.9	46.4	44.8	48.1	49.3	50.7	47.2	42.1	34.7	44.8	42.3	47.3
Llama-3.1-70b	60.5	51.2	41.1	42.2	33.6	45.7	41.5	49.9	60.5	53.7	45.9	39.0	28.6	45.5	40.3	50.8
GPT-4o-mini	51.5	50.1	43.1	43.2	38.7	45.3	43.2	47.5	51.5	45.6	37.8	35.0	26.3	39.2	35.2	43.3
Gemini-1.5 Pro	45.6	43.4	32.8	39.2	42.2	40.6	39.9	41.4	45.6	35.1	31.9	30.0	31.5	34.8	32.6	37.0
Gemini-1.5 Flash	36.8	32.0	34.3	27.2	26.8	31.4	29.8	33.1	36.8	30.7	25.8	27.7	20.5	28.3	25.9	30.7
Llama-3.1-8b	39.1	27.4	28.7	9.8	10.0	23.0	18.0	28.1	39.1	34.7	21.0	6.6	1.1	20.5	13.5	27.4

詳細な性能分析

Information amount has a greater effect on LCLM performance than context length

LMs	Constant Information Amount (2k)								Constant Information Density (50%)							
	4k	8k	16k	32k	64k	Avg.	wAvg. (inc)	wAvg. (dec)	4k	8k	16k	32k	64k	Avg.	wAvg. (inc)	wAvg. (dec)
Llama-3.1-405b	64.0	61.0	57.5	55.1	44.8	56.5	53.5	59.4	64.0	59.4	55.1	41.4	30.2	50.0	44.3	55.7
GPT-4o	62.5	52.2	53.6	59.4	52.0	55.9	55.0	56.8	62.5	57.5	45.3	43.3	37.5	49.2	45.0	53.5
o1-mini	63.5	55.8	48.8	47.6	40.3	51.2	47.5	54.8	63.5	51.2	45.8	34.4	29.2	44.8	39.1	50.5
Claude-3.5-Sonnet	49.3	52.2	46.4	42.3	41.9	46.4	44.8	48.1	49.3	50.7	47.2	42.1	34.7	44.8	42.3	47.3
Llama-3.1-70b	60.5	51.2	41.1	42.2	33.6	45.7	41.5	49.9	60.5	53.7	45.9	39.0	28.6	45.5	40.3	50.8
GPT-4o-mini	51.5	50.1	43.1	43.2	38.7	45.3	43.2	47.5	51.5	45.6	37.8	35.0	26.3	39.2	35.2	43.3
Gemini-1.5 Pro	45.6	43.4	32.8	39.2	42.2	40.6	39.9	41.4	45.6	35.1	31.9	30.0	31.5	34.8	32.6	37.0
Gemini-1.5 Flash	36.8	32.0	34.3	27.2	26.8	31.4	29.8	33.1	36.8	30.7	25.8	27.7	20.5	28.3	25.9	30.7
Llama-3.1-8b	39.1	27.4	28.7	9.8	10.0	23.0	18.0	28.1	39.1	34.7	21.0	6.6	1.1	20.5	13.5	27.4



詳細な性能分析

Llama-3.1-405b が短い文脈では最高精度

LMs	Constant Information Amount (2k)								Constant Information Density (50%)							
	4k	8k	16k	32k	64k	Avg.	wAvg. (inc)	wAvg. (dec)	4k	8k	16k	32k	64k	Avg.	wAvg. (inc)	wAvg. (dec)
Llama-3.1-405b	64.0	61.0	57.5	55.1	44.8	56.5	53.5	59.4	64.0	59.4	55.1	41.4	30.2	50.0	44.3	55.7
GPT-4o	62.5	52.2	53.6	59.4	52.0	55.9	55.0	56.8	62.5	57.5	45.3	43.3	37.5	49.2	45.0	53.5
o1-mini	63.5	55.8	48.8	47.6	40.3	51.2	47.5	54.8	63.5	51.2	45.8	34.4	29.2	44.8	39.1	50.5
Claude-3.5-Sonnet	49.3	52.2	46.4	42.3	41.9	46.4	44.8	48.1	49.3	50.7	47.2	42.1	34.7	44.8	42.3	47.3
Llama-3.1-70b	60.5	51.2	41.1	42.2	33.6	45.7	41.5	49.9	60.5	53.7	45.9	39.0	28.6	45.5	40.3	50.8
GPT-4o-mini	51.5	50.1	43.1	43.2	38.7	45.3	43.2	47.5	51.5	45.6	37.8	35.0	26.3	39.2	35.2	43.3
Gemini-1.5 Pro	45.6	43.4	32.8	39.2	42.2	40.6	39.9	41.4	45.6	35.1	31.9	30.0	31.5	34.8	32.6	37.0
Gemini-1.5 Flash	36.8	32.0	34.3	27.2	26.8	31.4	29.8	33.1	36.8	30.7	25.8	27.7	20.5	28.3	25.9	30.7
Llama-3.1-8b	39.1	27.4	28.7	9.8	10.0	23.0	18.0	28.1	39.1	34.7	21.0	6.6	1.1	20.5	13.5	27.4

詳細な性能分析

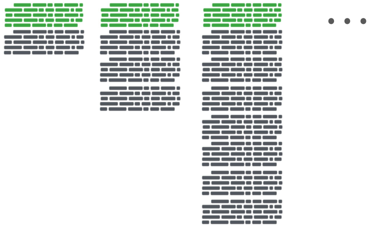
長文脈だと GPT-4o が支配的

LMs	Constant Information Amount (2k)								Constant Information Density (50%)							
	4k	8k	16k	32k	64k	Avg.	wAvg. (inc)	wAvg. (dec)	4k	8k	16k	32k	64k	Avg.	wAvg. (inc)	wAvg. (dec)
Llama-3.1-405b	64.0	61.0	57.5	55.1	44.8	56.5	53.5	59.4	64.0	59.4	55.1	41.4	30.2	50.0	44.3	55.7
GPT-4o	62.5	52.2	53.6	59.4	52.0	55.9	55.0	56.8	62.5	57.5	45.3	43.3	37.5	49.2	45.0	53.5
o1-mini	63.5	55.8	48.8	47.6	40.3	51.2	47.5	54.8	63.5	51.2	45.8	34.4	29.2	44.8	39.1	50.5
Claude-3.5-Sonnet	49.3	52.2	46.4	42.3	41.9	46.4	44.8	48.1	49.3	50.7	47.2	42.1	34.7	44.8	42.3	47.3
Llama-3.1-70b	60.5	51.2	41.1	42.2	33.6	45.7	41.5	49.9	60.5	53.7	45.9	39.0	28.6	45.5	40.3	50.8
GPT-4o-mini	51.5	50.1	43.1	43.2	38.7	45.3	43.2	47.5	51.5	45.6	37.8	35.0	26.3	39.2	35.2	43.3
Gemini-1.5 Pro	45.6	43.4	32.8	39.2	42.2	40.6	39.9	41.4	45.6	35.1	31.9	30.0	31.5	34.8	32.6	37.0
Gemini-1.5 Flash	36.8	32.0	34.3	27.2	26.8	31.4	29.8	33.1	36.8	30.7	25.8	27.7	20.5	28.3	25.9	30.7
Llama-3.1-8b	39.1	27.4	28.7	9.8	10.0	23.0	18.0	28.1	39.1	34.7	21.0	6.6	1.1	20.5	13.5	27.4

詳細な性能分析

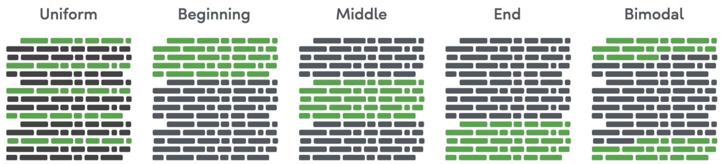
モデルサイズ が大きいほど、長文での劣化が緩やか

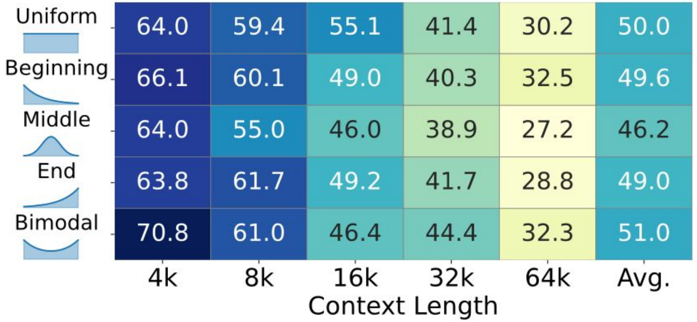
LMs	Constant Information Amount (2k)								Constant Information Density (50%)							
	4k	8k	16k	32k	64k	Avg.	wAvg. (inc)	wAvg. (dec)	4k	8k	16k	32k	64k	Avg.	wAvg. (inc)	wAvg. (dec)
Llama-3.1-405b	64.0	61.0	57.5	55.1	44.8	56.5	53.5	59.4	64.0	59.4	55.1	41.4	30.2	50.0	44.3	55.7
GPT-4o	62.5	52.2	53.6	59.4	52.0	55.9	55.0	56.8	62.5	57.5	45.3	43.3	37.5	49.2	45.0	53.5
o1-mini	63.5	55.8	48.8	47.6	40.3	51.2	47.5	54.8	63.5	51.2	45.8	34.4	29.2	44.8	39.1	50.5
Claude-3.5-Sonnet	49.3	52.2	46.4	42.3	41.9	46.4	44.8	48.1	49.3	50.7	47.2	42.1	34.7	44.8	42.3	47.3
Llama-3.1-70b	60.5	51.2	41.1	42.2	33.6	45.7	41.5	49.9	60.5	53.7	45.9	39.0	28.6	45.5	40.3	50.8
GPT-4o-mini	51.5	50.1	43.1	43.2	38.7	45.3	43.2	47.5	51.5	45.6	37.8	35.0	26.3	39.2	35.2	43.3
Gemini-1.5 Pro	45.6	43.4	32.8	39.2	42.2	40.6	39.9	41.4	45.6	35.1	31.9	30.0	31.5	34.8	32.6	37.0
Gemini-1.5 Flash	36.8	32.0	34.3	27.2	26.8	31.4	29.8	33.1	36.8	30.7	25.8	27.7	20.5	28.3	25.9	30.7
Llama-3.1-8b	39.1	27.4	28.7	9.8	10.0	23.0	18.0	28.1	39.1	34.7	21.0	6.6	1.1	20.5	13.5	27.4



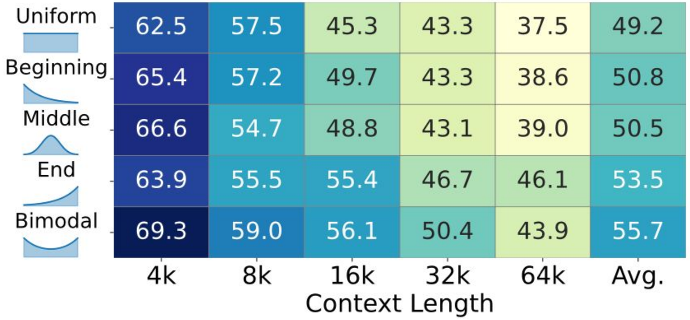
情報ポジションの分析

- モデルごとに好む場所は異なる
- Llama-3.1-405bはGPT-4oよりも情報ポジションに関して頑健である





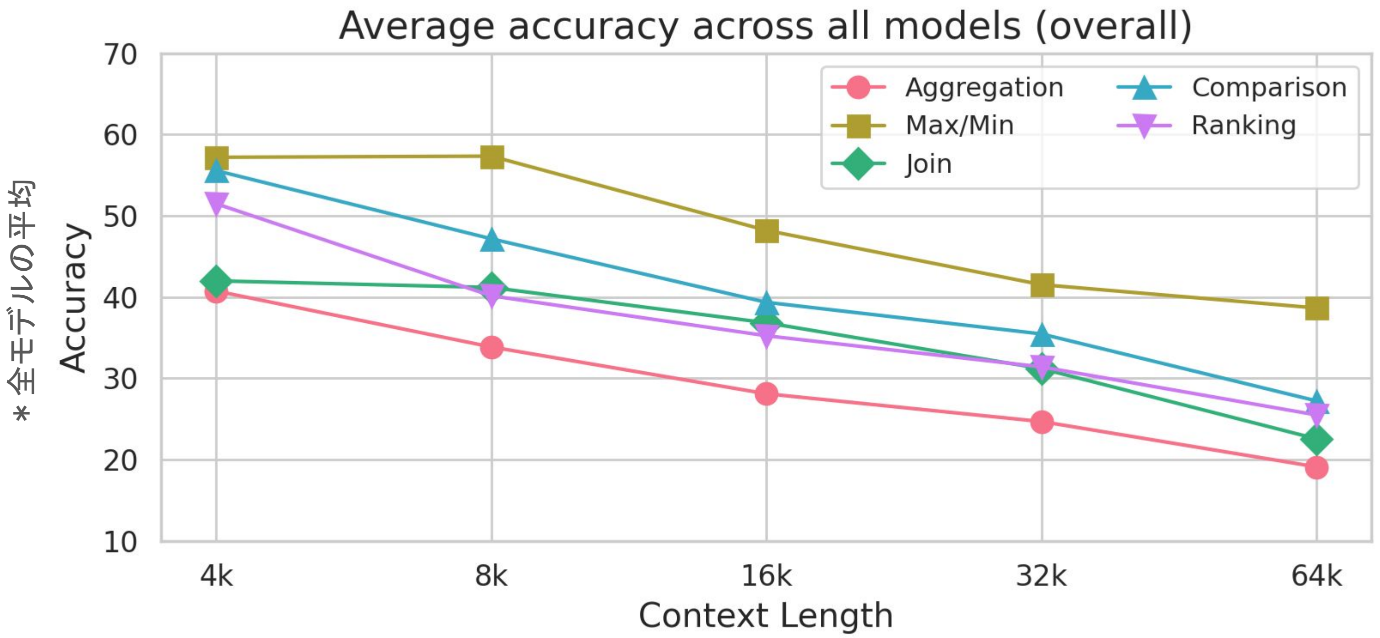
(a) Llama-3.1-405b



(b) GPT-4o

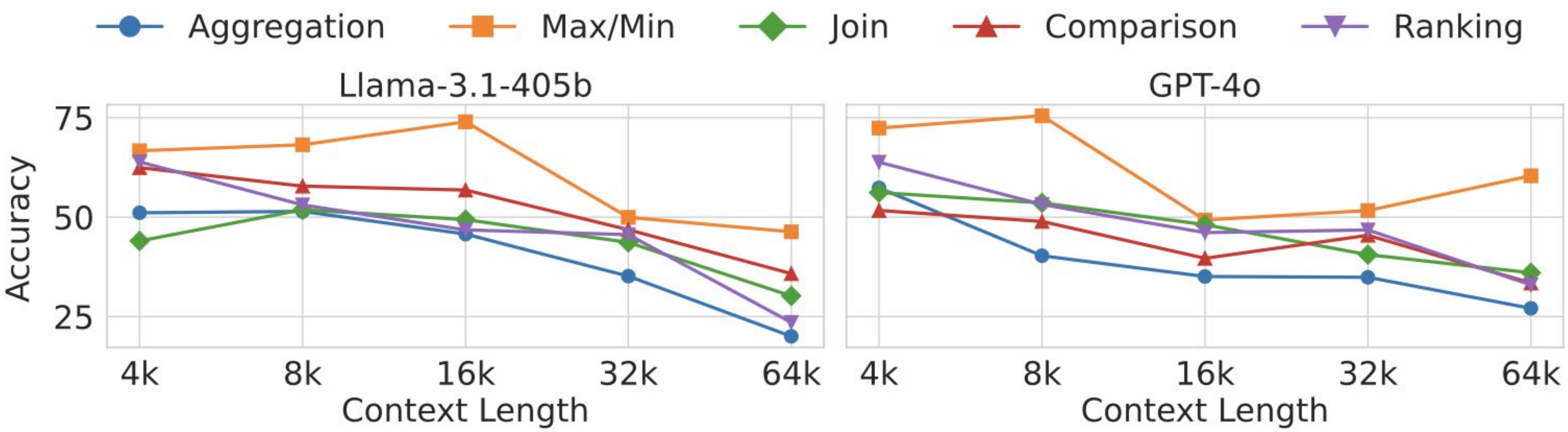
クエリタイプ分析

Aggregationは難しく、Max/Minは比較的簡単



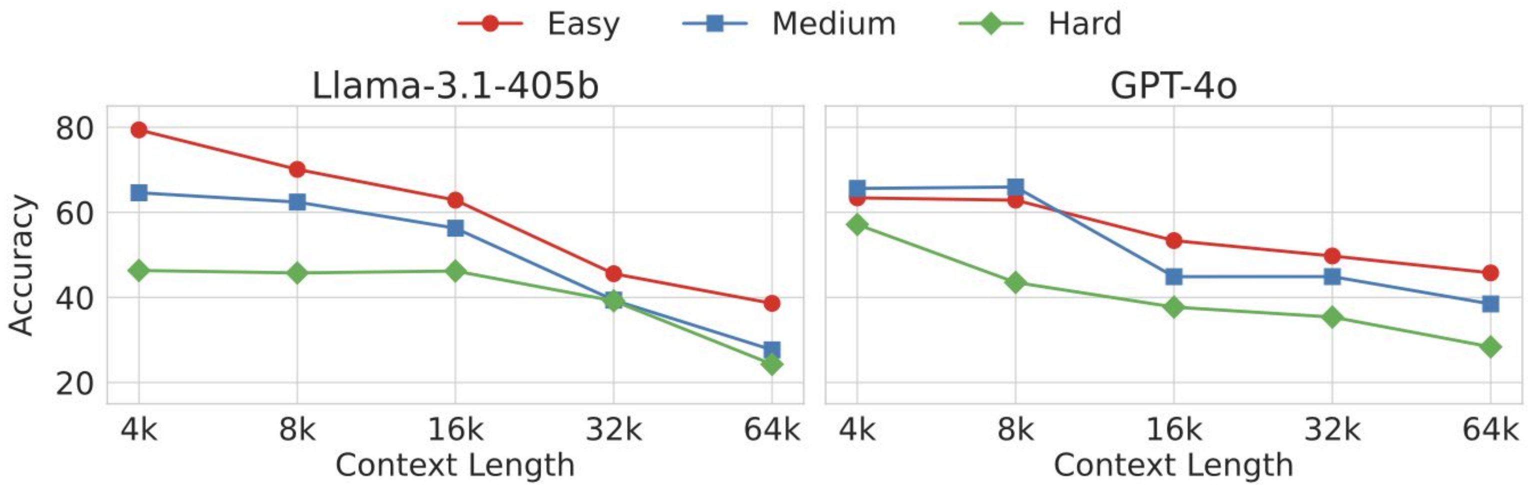
Query Type Affects Performance

- Max/Min queries are easier than Aggregation queries
  - Extreme value tasks are simpler and do not scale in complexity with longer context lengths.



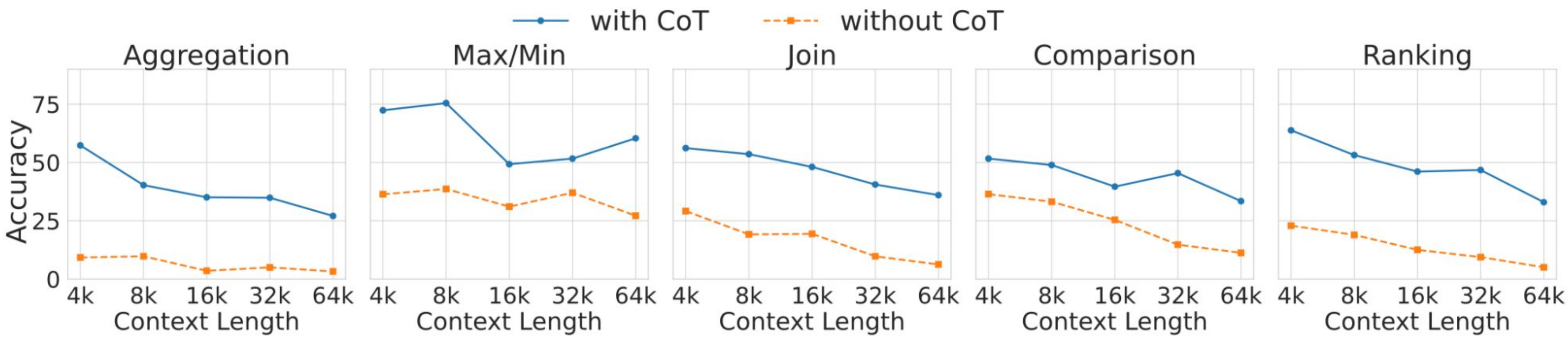
クエリ複雑さの影響

- 難易度順 : Easy > Medium > Hard
- Hard クエリは長文になるほど正答率が急落



Chain-of-Thought (CoT) の効果

- Aggregation クエリ:
  - CoTなしではほぼ解けない
  - 情報合成にCoTは重要な役割
- Max/Min クエリ:
  - 簡単なためCoTへの依存度は低い
  - 文脈長の影響が小さい



得られた知見のおさらい

- 文脈長よりも情報量の増加が性能低下を招く
- LCLMsであっても複雑な情報合成はうまく解けない
- CoT プロンプトは複雑推論に必須

モデルの分析：

- Llama 3.1-405 B: 短文と位置頑健性で優位
- GPT-4o: 長文と一部タスクで優位

HoloBench is available at: https://github.com/megagonlabs/holobench

おまけ（新しいモデルの追加）

LCLMs	Constant Information Amount (2k)								Constant Information Density (50%)							
	4k	8k	16k	32k	64k	Avg.	wAvg. (inc)	wAvg. (dec)	4k	8k	16k	32k	64k	Avg.	wAvg. (inc)	wAvg. (dec)
Llama-3.1-405b	64.0	61.0	57.5	55.1	44.8	56.5	53.5	59.4	64.0	59.4	55.1	41.4	30.2	50.0	44.3	55.7
GPT-4o	62.5	52.2	53.6	59.4	52.0	55.9	55.0	56.8	62.5	57.5	45.3	43.3	37.5	49.2	45.0	53.5
o1-mini	63.5	55.8	48.8	47.6	40.3	51.2	47.5	54.8	63.5	51.2	45.8	34.4	29.2	44.8	39.1	50.5
Claude-3.5-Sonnet	49.3	52.2	46.4	42.3	41.9	46.4	44.8	48.1	49.3	50.7	47.2	42.1	34.7	44.8	42.3	47.3
Llama-3.1-70b	60.5	51.2	41.1	42.2	33.6	45.7	41.5	49.9	60.5	53.7	45.9	39.0	28.6	45.5	40.3	50.8
GPT-4o-mini	51.5	50.1	43.1	43.2	38.7	45.3	43.2	47.5	51.5	45.6	37.8	35.0	26.3	39.2	35.2	43.3
Gemini-1.5 Pro	45.6	43.4	32.8	39.2	42.2	40.6	39.9	41.4	45.6	35.1	31.9	30.0	31.5	34.8	32.6	37.0
Gemini-1.5 Flash	36.8	32.0	34.3	27.2	26.8	31.4	29.8	33.1	36.8	30.7	25.8	27.7	20.5	28.3	25.9	30.7
Llama-3.1-8b	39.1	27.4	28.7	9.8	10.0	23.0	18.0	28.1	39.1	34.7	21.0	6.6	1.1	20.5	13.5	27.4
Gemini-2.0 Flash	66.2	62.3	59.1	57.3	53.0	59.6	57.5	61.7	66.2	60.1	57.0	52.5	48.2	56.8	53.9	59.7
Gemini-2.0 Flash Thinking	69.5	54.2	59.5	59.6	43.0	57.2	54.0	60.3	69.5	55.3	51.9	43.2	30.3	50.1	44.0	56.1
o3-mini	68.6	62.6	51.6	46.8	42.7	54.5	50.0	59.0	68.6	58.0	47.2	33.6	22.4	46.0	38.2	53.8
DeepSeek-V3	59.5	51.7	50.8	47.3	53.6	52.6	51.5	53.7	59.5	56.9	52.5	45.8	36.0	50.1	46.2	54.0
DeepSeek-R1	51.1	41.8	34.1	24.1	36.4	37.5	34.4	40.7	51.1	60.4	31.9	24.6	11.3	35.9	28.2	43.6

Reasoning models は短文脈では強力だが、長文脈での劣化が顕著

Overthinking の問題？

Appendix

Template Examples

Table/Database	Template
customers/store_1	In {city}, {country}, {full_name} can be reached at {phone} for assistance from {support_rep_employee_name}.
airports/flight_4	The {name} is located in {city}, {country} at an elevation of {elevation} feet.
Player/soccer_1	With a height of {height} cm, {player_name} was born on {birthday} and weighs {weight} lb.

Table 4: Templates for text generation. {placeholder} indicate column names that can be replaced with actual values from the tables.

Database operations we are targeting

Query Type	Natural language / SQL queries
Aggregation	What is the total number of unique instructors? `SELECT count(DISTINCT instructor_name) FROM advisor;`
Max/Min	What is the maximum capacity of any classroom? `SELECT MAX(capacity) FROM classroom;`
Join	List the course titles and the corresponding buildings. `SELECT course.title, section.building FROM course INNER JOIN section ON course.title = section.course_title;`
Comparison	Find the buildings which have rooms with capacity more than 50. `SELECT DISTINCT building FROM classroom WHERE capacity > 50;`
Ranking	What are the titles of the top 5 posts with the highest popularity? `SELECT Title FROM posts ORDER BY ViewCount DESC LIMIT 5;`

Table 2: Categorization of query types with example natural language and SQL queries.